Question 5: Short Answer — 4 points

A	For the correct answer: *Tetrahedral*	**Point 01**
B	For the correct answer and a valid justification: *Agree. Compound* Y *has a larger, more polarizable electron cloud because the* Si *atom has more occupied electron shells than the* C *atom, giving compound* Y *stronger London dispersion forces and a higher boiling point than compound* X.	**Point 02**
C	For the correct answer and a valid justification. Examples of acceptable responses may include the following: • *Compound* X. *Compound* X *has weaker intermolecular forces than compound* Y, *so molecules of* X *are more likely to be in the gas phase at* 82°C *and would therefore have a higher vapor pressure.* • *Compound* X. *At* 82°C, *compound* X *has reached its boiling point, but compound* Y *has not. Therefore, the proportion of* X *molecules in the vapor phase would be much greater than that of compound* Y, *giving compound* X *the higher vapor pressure.*	**Point 03**
D	For the correct calculated value: $PV = nRT$ $n = \frac{PV}{RT} = \frac{(2.30\text{ atm})(12.5\text{ L})}{(0.08206\frac{\text{L}\cdot\text{atm}}{\text{K}\cdot\text{mol}})(471\text{K})} = 0.744\text{ mol}$	**Point 04**

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